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Report 1

Data Analysis with Python

Predictive Maintenance in Nuclear Power Plant Subsystems

Assignment 01

Dataset: Nuclear Power Plant Alarm Dataset (IEEE DataPort, DOI: 10.21227/g2fa-9y43)

1. Introduction

This report is basically the data analysis part of Assignment 01, under the theme Predictive Maintenance in Nuclear Power Plant Subsystems. The brief wanted one real world dataset that would be used in both the data analysis study and the machine learning study, so the same dataset, and also the same cleaned version, stays consistent all the way into Report 2. Here the dataset used is the “Nuclear Power Plant Alarm Dataset”, which is published on IEEE DataPort by Gianluca Manca (ABB Corporate Research Center, Germany) and Franz C. Kunze (Ruhr University). It was chosen because it contains authentic sensor–alarm behaviour, but from a simulated nuclear plant, and that is pretty much the structured signal type a predictive maintenance system would keep an eye on in real settings. In other words, instead of waiting for a subsystem’s own alarm to trigger, predictive maintenance tries to detect the early pattern of readings that tends to show up before the alarm.

So the goal of this report is to show how the dataset was checked, refined and explored at the subsystem level , and then explain what that exploration suggests about where most alarm activity seems to cluster in the plant. It also looks at which signals show a behaviour that a predictive model could feasibly learn. The modelling step itself, meaning training and evaluating a classifier that predicts a subsystem alarm from other sensor states and it is built directly from the findings developed in this report.

2. Dataset Description and Justification

The dataset is derived from a larger open dataset of simulated nuclear power plant accidents produced with PCTRAN, a simulator used by the International Atomic Energy Agency for its annual Advanced NPP Simulation Workshop. The original dataset (Qi et al., 2022, Scientific Data) provided 1,211 simulated accident runs across a number of fault classes but did not include alarm information. Manca and Kunze generated alarm time series from that simulation data by setting high/low thresholds for each process variable, based on a trade-off between missed alarm rate (MAR) and false alarm rate (FAR), producing a benchmark dataset intended for alarm-management and alarm-flood research.

For this assignment, one sample run was used together with its corresponding alarm-threshold profiles.

- Alarm data (alarm_df): 302 time steps recorded at 10-second intervals, across 193 columns — one TIME column plus 192 paired HI/LO binary alarm flags (0 = not triggered, 1 = triggered) covering 96 process sensors spread across the plant's subsystems.
- Alarm threshold data (thresholds_df): 10 rows, one per alpha (MAR/FAR trade-off) configuration, across 97 columns describing the high- and low-alarm limits used to generate the alarm data for each sensor.

This dataset suits a predictive-maintenance framing well. it is fully numeric and tabular, it covers multiple identifiable plant subsystems rather than a single sensor and because a subsystem alarm can be treated as an early indicator of abnormal operating conditions a model that anticipates a subsystem alarm from the state of related sensors is functionally an early-warning tool for that subsystem, which is the practical goal of predictive maintenance in a process-safety setting.

3. Data Preprocessing and Cleaning

Before any analysis, the dataset first had to be extracted from the two 7z archives it is distributed in (NPP Alarm Dataset.7z and NPP Alarm Thresholds.7z), using the py7zr library. The relevant CSV files were then found and pulled into two pandas DataFrames, alarm_df for the alarm time series and thresholds_df for the threshold profiles.

3.1 Missing values and duplicates

Both DataFrames were looked over for missing values using isnull().sum().sum, and neither one had anything in it: 0 missing values in the alarm dataset, and 0 in the thresholds dataset. A duplicate check with duplicated().sum() as well was clean too 0 duplicate rows in either table. That matches the idea that the data was produced programmatically by a simulator rather than gathered from messy physical sensors, but we didn't just assume it. It was actually verified, which meant no imputation, or any row dropping, was necessary before analysis.

3.2 Shape after cleaning

The final working shapes were 302 rows x 193 columns for the alarm dataset and 10 rows x 97 columns for the thresholds dataset. Since the data was already numeric, binary/continuous and also complete, the cleaning stage kinda mainly helped to double check data quality more than it did any true remediation, so it basically left the rest of the analysis free to concentrate on subsystem level exploration, kind of like drilling down further instead of fixing the obvious.

4. Subsystem Grouping Methodology Note

The alarm dataset's 96 sensors are provided as individual tags (P, TAVG, THA, VOL, MDBR, and so on) with no subsystem labels attached in the source documentation. To analyse the data at a predictive-maintenance-relevant level i.e. by physical subsystem rather than by 96 separate tags sensor tags were grouped into four broad subsystems using standard NPP naming conventions. Primary Coolant System, Secondary Steam Cycle, Control Rod Drive Mechanism, and Emergency Diesel Generators.

Methodological caveat- this subsystem mapping is kind of a working assumption, we built it for this analysis not an official grouping published with the dataset. It is based on conventional NPP tag naming patterns, like pressure/temperature tags we map them to the Primary Coolant System, and rod-drive tags we map them to the Control

Rod Drive Mechanism. So you should read it as a reasonable analytical lens for organising the sensors, not as a verified engineering diagram of the simulated plant. Also when a subsystem shows zero alarm activity below , this can mean either there were really no alarms in that area, or it can just mean that no sensor tag in this sample matched the subsystem’s naming pattern.

5. Exploratory Data Analysis

With that grouping in place, the EDA was organised around four questions relevant to predictive maintenance: how much do alarm thresholds vary across configurations, which subsystems generate the most alarm activity, which individual sensors alarm most often, and how do alarm states change over time.

5.1 Descriptive Statistics of Alarm Thresholds

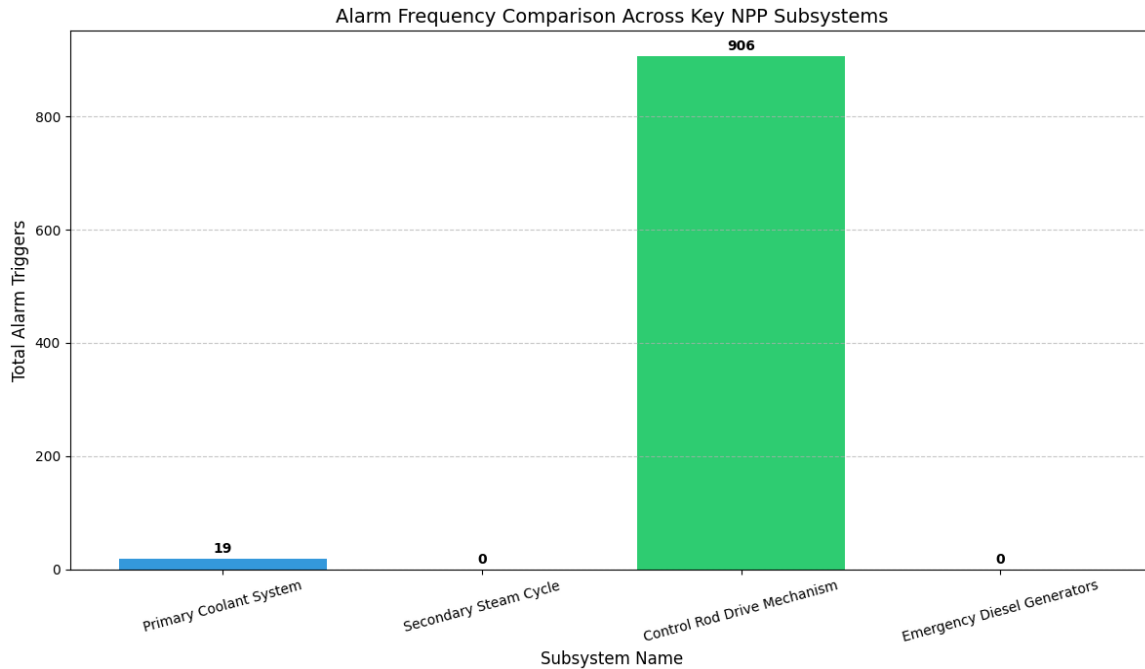
The thresholds dataset defines, for each of the 10 alpha configurations, the high/low alarm limit chosen for every sensor. Looking at five representative variables from the Primary Coolant System pressure (P) and four temperature measurements (TAVG, THA, THB, TCA) gives a first sense of how much these limits vary:

Statistic	P	TAVG	THA	THB	TCA
Count	10	10	10	10	10
Mean	30.71	61.19	63.40	63.40	59.38
Std. Dev.	64.34	128.19	132.91	132.91	124.74
Min	0.00	0.00	0.00	0.00	0.00
25%	0.01	0.15	0.14	0.14	0.00
50% (Median)	0.34	0.54	0.55	0.55	0.36
75%	0.54	0.82	0.83	0.83	0.55
Max	157.12	309.91	324.81	324.80	300.55

The average pressure threshold hangs around 30.7 units, however the standard deviation 64.3 is way more than twice the mean, and this same strange pattern kind of repeats across the temperature variables too. Take THA, for instance , it shows a mean of 63.4 but the standard deviation jumps to 132.9, and the maximum lands at 324.8. As for the 10 alpha configurations, they really don’t settle into kind of similar threshold values; instead, depending on how the missed-alarm/false-alarm balance is weighted the selected limits swing from 0 up into the low hundreds. For a predictive maintenance system this isn’t just a detail. It matters directly: a model trained on alarms produced under one particular threshold setup should not be treated as if it will just port over to a plant running a differently tuned alarm mechanism, because what counts as “abnormal” is redefined each time the configuration changes.

5.2 Alarm Activity by Subsystem

Using the subsystem grouping described in Section 4, six sensor tags mapped to the Primary Coolant System, three to the Secondary Steam Cycle, and three to the Control Rod Drive Mechanism no tags in this sample matched the Emergency Diesel Generator naming pattern.

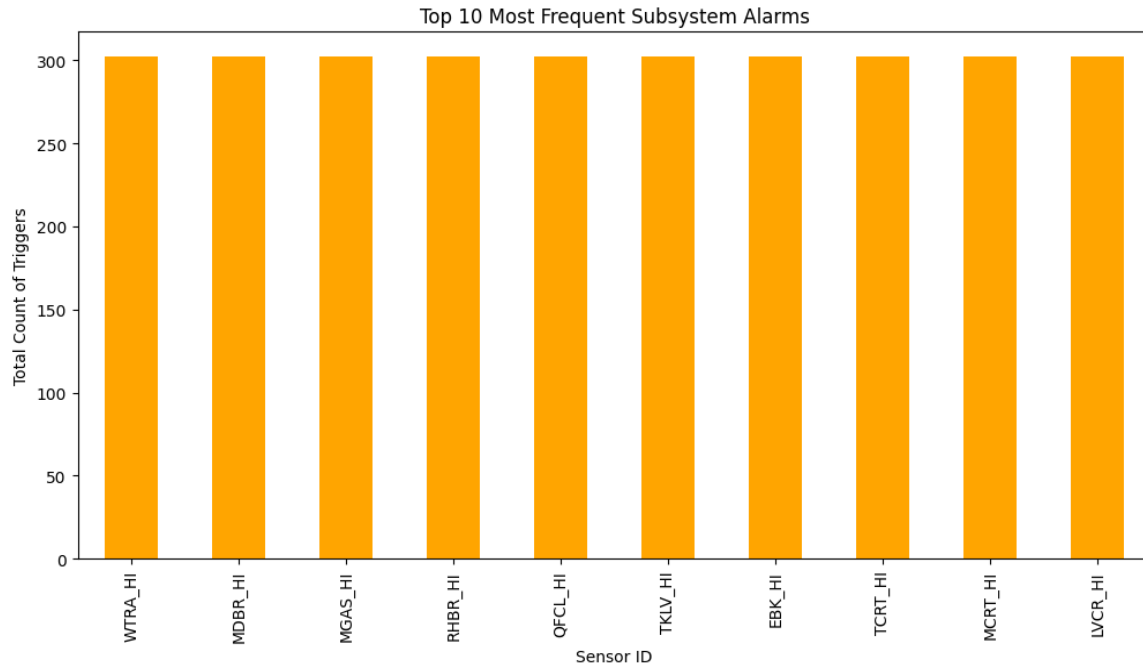


Subsystem (working grouping)	Sensors Mapped	Total Alarm Triggers (302 timesteps)
Primary Coolant System	6	19
Secondary Steam Cycle	3	0
Control Rod Drive Mechanism	3	906
Emergency Diesel Generators	0 (no matching columns)	0

The Control Rod Drive Mechanism dominates the picture with 906 cumulative alarm triggers, roughly 48 times the Primary Coolant System's 19, while the Secondary Steam Cycle records none at all in this sample. From a predictive-maintenance standpoint this immediately flags the Control Rod Drive Mechanism as the subsystem most in need of monitoring attention but Section 5.3 shows that this concentration is driven by a small number of sensors being continuously active rather than by frequent, distinct fault events, which changes how that finding should be acted on.

5.3 Most Frequently Triggered Alarms

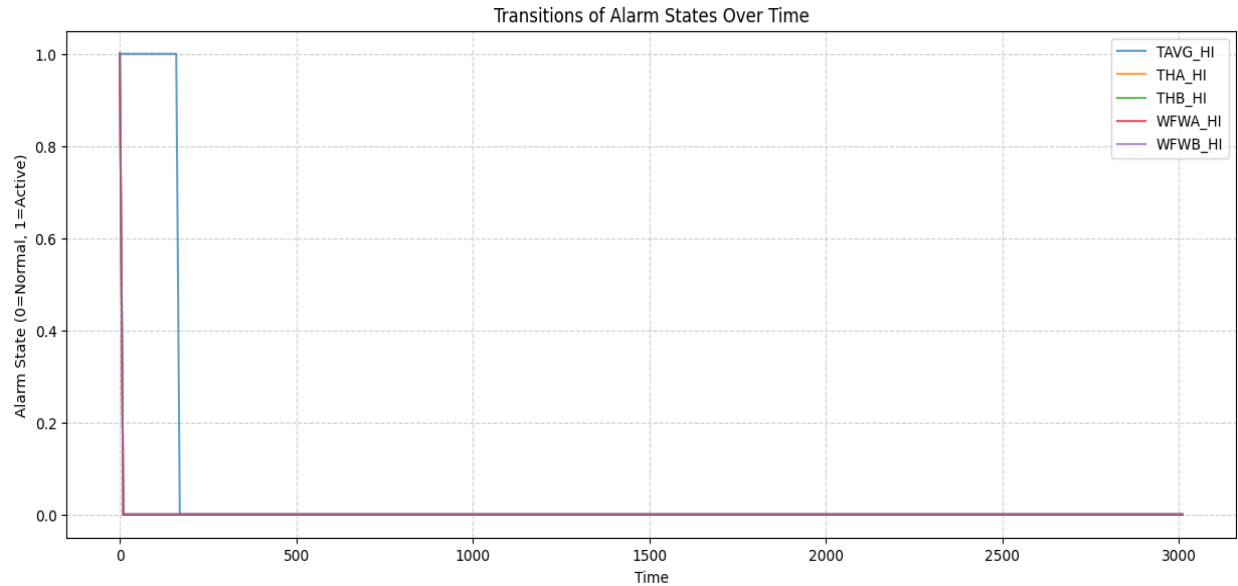
Summing each alarm column across all 302 time steps and ranking the top 10 confirms the pattern noticed above:



All of the top 10 alarms (WTRA_HI, MDBR_HI, MGAS_HI, RHBR_HI, QFCL_HI, TKLV_HI, EBK_HI, TCRT_HI, MCRT_HI, LVCR_HI) are active for all 302 time steps in this sample, the count ends up 302 for each one. A few of them, (MDBR_HI, MGAS_HI, MCRT_HI) are tied to the Control Rod Drive Mechanism grouping, and that's why that subsystem's high trigger total shows up in Section 5.2. In other words these are "stuck on" alarms during this run, not 906 individual maintenance worthy incidents. For predictive maintenance, an alarm that stays on permanently just doesn't add much forward-looking information, because there's no useful transition to forecast. A maintenance model needs the sensors that actually shift between normal and abnormal, and that's exactly the kind of pattern the next section is looking at.

5.4 Temporal Behaviour of Alarm States

Because most alarm columns in this run are always 0 or always 1 (basically), the columns that had more than one unique value were separated, just to track those state changes the predictive maintenance model really needs to learn from. When we plot the first five TAVG_HI, THA_HI, THB_HI, WFWA_HI, WFWB_HI it turns out they're not flat at all, you can actually see dynamic behaviour emerge:



Most of the alarms that got plotted (WFWA_HI, WFWB_HI, THA_HI, THB_HI) are only active for the first one or two time steps, then they clear and kinda stay quiet. But TAVG_HI (average coolant temperature, high) which is part of the Primary Coolant System grouping, it stays active for roughly the first 150 seconds then it goes back to normal and it remains there for the rest of the run. That sustained, then cleared pattern is exactly the sort of thing a predictive maintenance model can learn from, because there really is this normal-then-abnormal-then-normal shift you'd want to predict, so that's why TAVG_HI was carried forward as the prediction target for the Primary Coolant System in Report 2.

6. Discussion of Findings

Bringing the subsystem-level analysis together, several points are directly relevant to a predictive-maintenance strategy for this simulated plant:

- Data quality supports reliable maintenance signals. Zero missing values and zero duplicates across both tables mean the patterns identified reflect the simulated process itself, not gaps in data collection – an important baseline before trusting any model built on top of it.
- Threshold configuration changes what counts as 'abnormal'. The wide variance in alarm thresholds across the 10 alpha configurations (Section 5.1) means a predictive-maintenance model is implicitly tied to the threshold regime it was trained under, and that dependency should be documented and monitored if the model is ever deployed against a differently configured alarm system.
- Alarm activity is concentrated in the Control Rod Drive Mechanism, but not evenly. That subsystem accounts for the large majority of raw alarm triggers (Section 5.2), yet Section 5.3 shows this is driven by a handful of continuously-active sensors rather than repeated distinct events so raw trigger counts alone would overstate how much genuine maintenance-relevant activity is happening there.
- The Primary Coolant System offers the clearest predictive-maintenance opportunity in this sample. TAVG_HI's genuine transition behaviour (Section 5.4) makes it the one alarm in this dataset that a model can meaningfully learn to anticipate, rather than merely memorise as constant.

all that put together kinda implies a practical two stage predictive maintenance workflow for a dataset like this: first you split out chronic, always on alarms from the ones that are actually dynamic , because the chronic ones usually mean fixed operating conditions or mis tuned thresholds, not something you can meaningfully act on in maintenance terms. Then you basically focus the modeling work on that dynamic subset, similar to how TAVG_HI was handled in Report 2. In that setup an early warning model can add real value way before the subsystem's own alarm triggers.

7. Conclusion

This data analysis exercise basically confirmed that the Nuclear Power Plant Alarm Dataset is clean , like really well-structured, and basically suitable for a predictive-maintenance study at the subsystem level too. When grouping the 96 individual sensors into four working subsystems, it became obvious that the alarm activity is heavily concentrated in the Control Rod Drive Mechanism, but then, on closer inspection, it seems that the “concentration” is mostly driven by continuously-active alarms rather than a bunch of frequent distinct fault events. The Primary Coolant System's TAVG_HI alarm stood out as the only variable here with a genuine normal-to-abnormal-to-normal transition , so it became the natural candidate for predictive modeling in practice. Overall these findings the subsystem groupings, the threshold sensitivity, and the identification of TAVG_HI as a learnable target directly set up the machine learning work carried out in Report 2 .

References

- [1] G. Manca and F. C. Kunze, "Nuclear Power Plant Alarm Dataset," IEEE DataPort, 2021 (updated 2024). doi: 10.21227/g2fa-9y43.
- [2] B. Qi, X. Xiao, J. Liang, L. C. Po, L. Zhang, and J. Tong, "An open time-series simulated dataset covering various accidents for nuclear power plants," Scientific Data, vol. 9, article 766, Dec. 2022. doi: 10.1038/s41597-022-01879-1.
- [3] L. C. Po, "Conceptual design of an accident prevention system for light water reactors using artificial neural network and high-speed simulator," Nuclear Technology, vol. 206, no. 3, pp. 505–513, Aug. 2019. doi: 10.1080/00295450.2019.1641877.